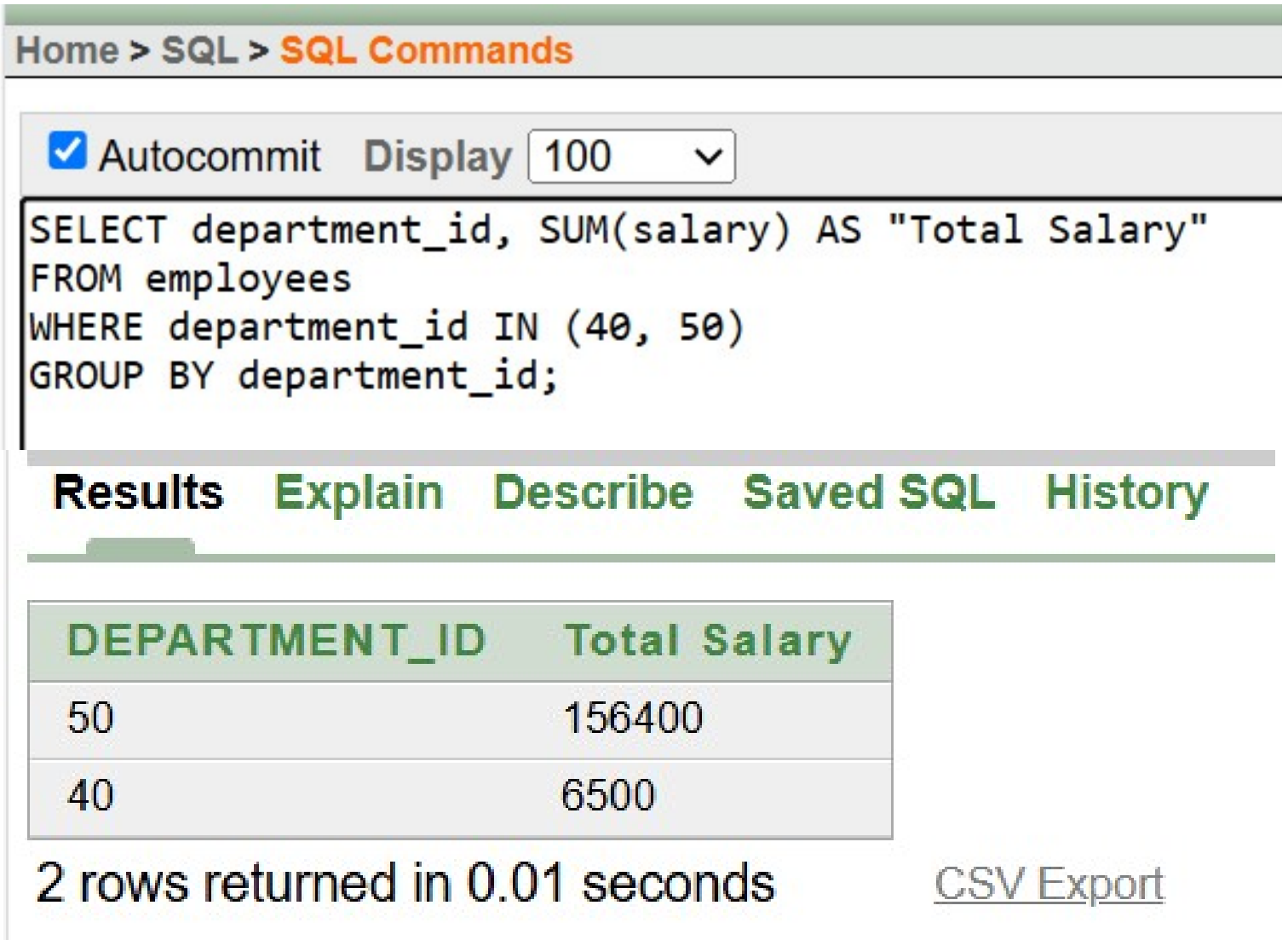
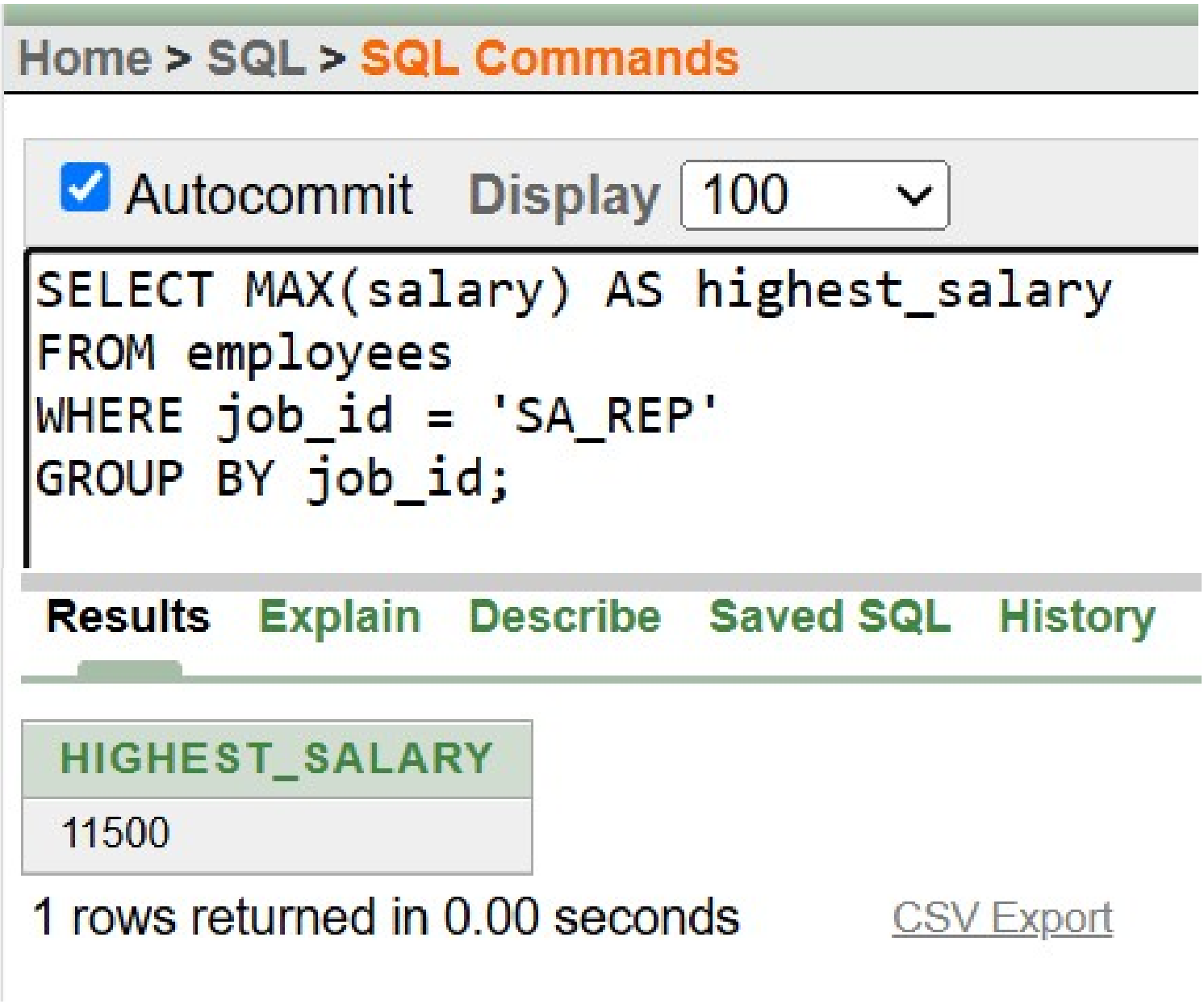
Assignment-9

(Group Functions)

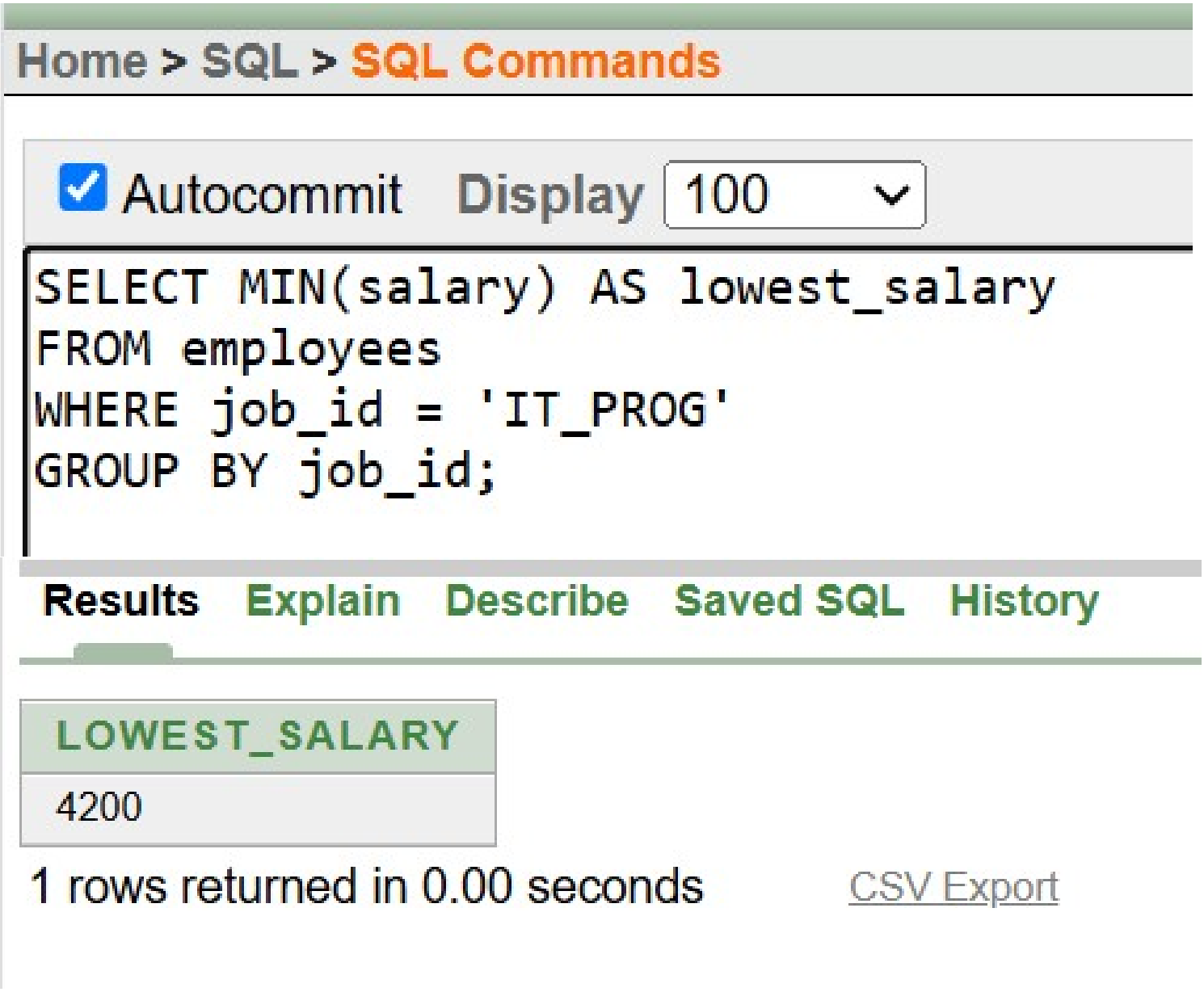
1. Write a query to display the sum total of salary for department 40 and 50.



2. **Display the highest salary for job sales representative.**



3. **Display the lowest salary for job IT programmer.**



4. **Select the average salary of all job_id.**

Home > SQL > SQL Commands

Autocommit Display 100

```
SELECT job_id, AVG(salary) AS average_salary
FROM employees
GROUP BY job_id;
```

Results Explain Describe Saved SQL History

JOB_ID	AVERAGE_SALARY
IT_PROG	5760
AC_MGR	12000
AC_ACCOUNT	8300
ST_MAN	7280
PU_MAN	11000
AD_ASST	4400
AD_VP	17000
SH_CLERK	3215
FI_ACCOUNT	7920
FI_MGR	12000
PU_CLERK	2780
SA_MAN	12200
MK_MAN	13000
PR_REP	10000
AD_PRES	24000
SA_REP	8350
MK_REP	6000
ST_CLERK	2785
HR_REP	6500

19 rows returned in 0.00 seconds CSV Export

5. **Display the number of employees present in each department.**

Home > SQL > SQL Commands

Autocommit Display 100

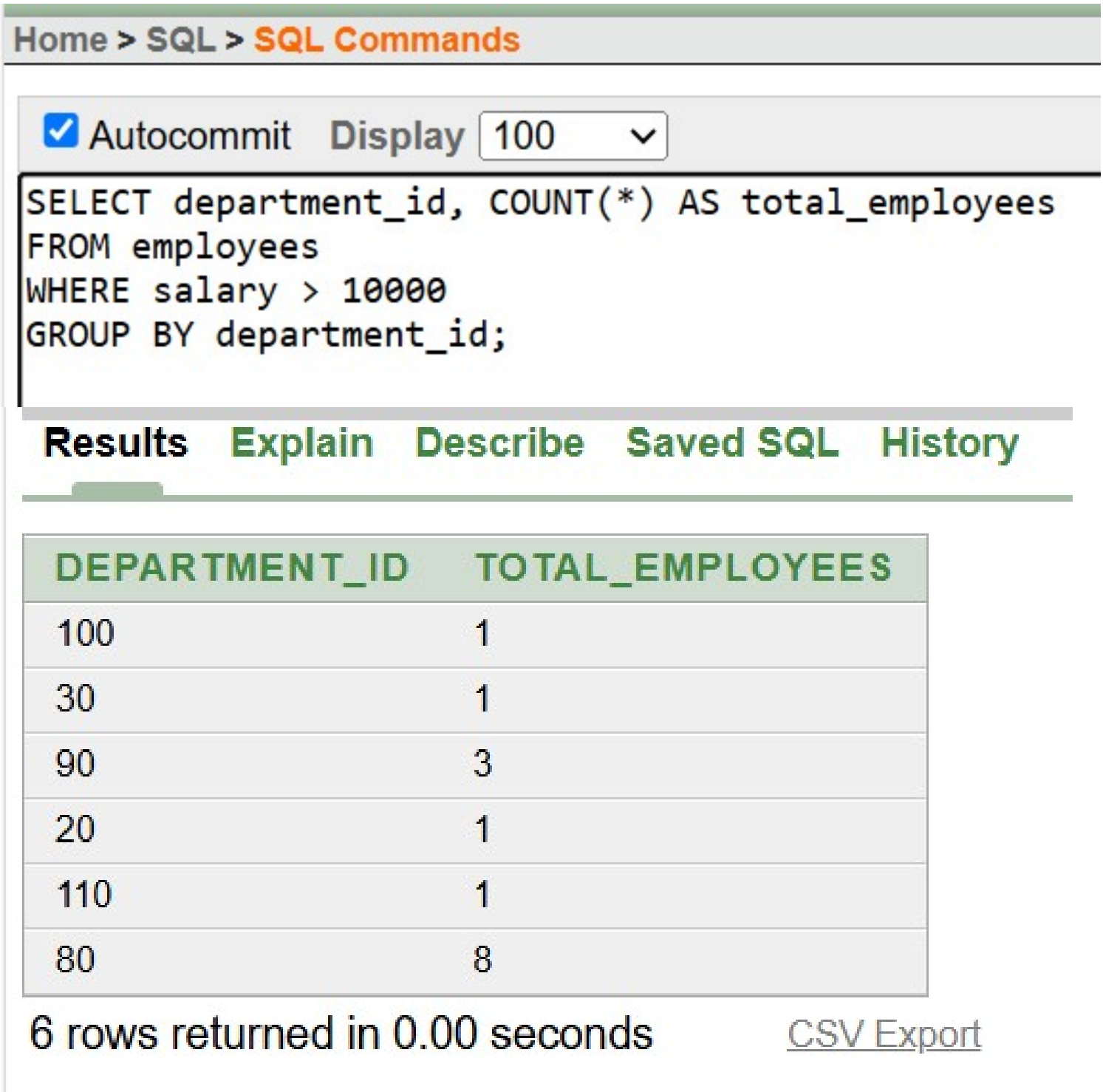
```
SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id;
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	TOTAL_EMPLOYEES
100	6
30	6
-	1
90	3
20	2
70	1
110	2
50	45
80	34
40	1
60	5
10	1

12 rows returned in 0.00 seconds CSV Export

6. **Display the number of employees present in each department whose salary is more than 10000.**



7. **Display department-wise maximum and minimum salary.**

Home > SQL > SQL Commands

Autocommit Display 100

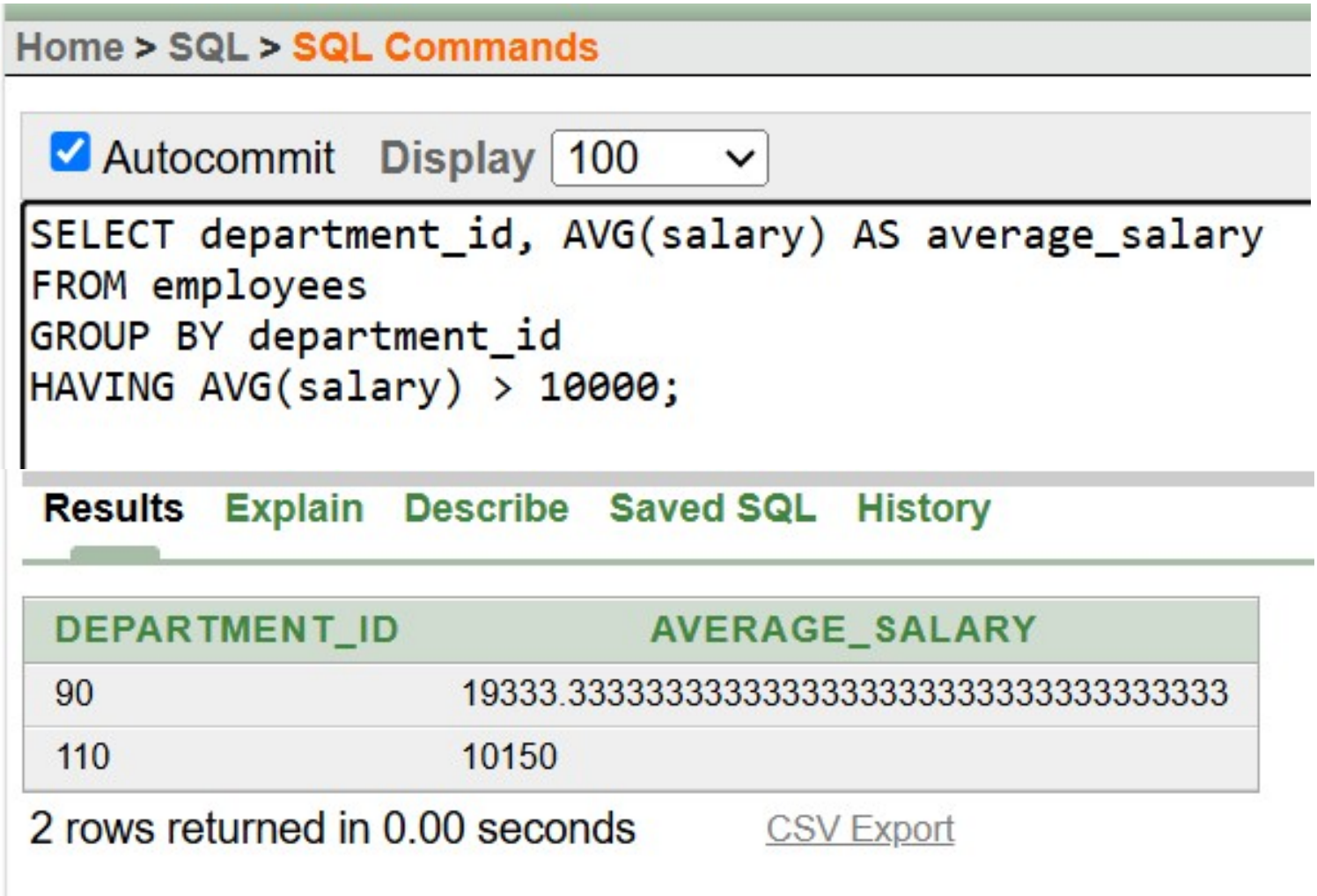
```
SELECT department_id,
       MAX(salary) AS maximum_salary,
       MIN(salary) AS minimum_salary
FROM employees
GROUP BY department_id;
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	MAXIMUM_SALARY	MINIMUM_SALARY
100	12000	6900
30	11000	2500
-	7000	7000
90	24000	17000
20	13000	6000
70	10000	10000
110	12000	8300
50	8200	2100
80	14000	6100
40	6500	6500
60	9000	4200
10	4400	4400

12 rows returned in 0.02 seconds CSV Export

8. **Display average salary of each department when average salary is more than 10000.**



9. **Write a query to Display number of managers working in each department.**

Home > SQL > SQL Commands

Autocommit Display 100

```
SELECT department_id,
       COUNT(DISTINCT manager_id) AS number_of_managers
FROM employees
GROUP BY department_id;
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	NUMBER_OF_MANAGERS
10	1
20	2
30	2
40	1
50	6
60	2
70	1
80	6
90	1
100	2
110	2
-	1

12 rows returned in 0.00 seconds CSV Export

10. **Display the employees department wise who have joined in 1997 & 1998 in ascending order of department_id.**

Home > SQL > SQL Commands

Autocommit Display 100

```
SELECT employee_id,
       first_name,
       last_name,
       department_id,
       hire_date
FROM employees
WHERE hire_date BETWEEN DATE '1997-01-01' AND DATE '1998-12-31'
ORDER BY department_id ASC, hire_date ASC;
```

Results Explain Describe Saved SQL History

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	HIRE_DATE
202	Pat	Fay	20	17-AUG-97
117	Sigal	Tobias	30	24-JUL-97
116	Shelli	Baida	30	24-DEC-97
118	Guy	Himuro	30	15-NOV-98
142	Curtis	Davies	50	29-JAN-97
131	James	Marlow	50	16-FEB-97
110	John	Chen	100	28-SEP-97
111	Ismael	Sciarra	100	30-SEP-97
112	Jose Manuel	Urman	100	07-MAR-98

51 rows returned in 0.00 seconds CSV Export

11. **Write a query to display the difference between highest and lowest salary in each department. Label the column “DIFFERENCE”.**

Home > SQL > SQL Commands

Autocommit Display 100

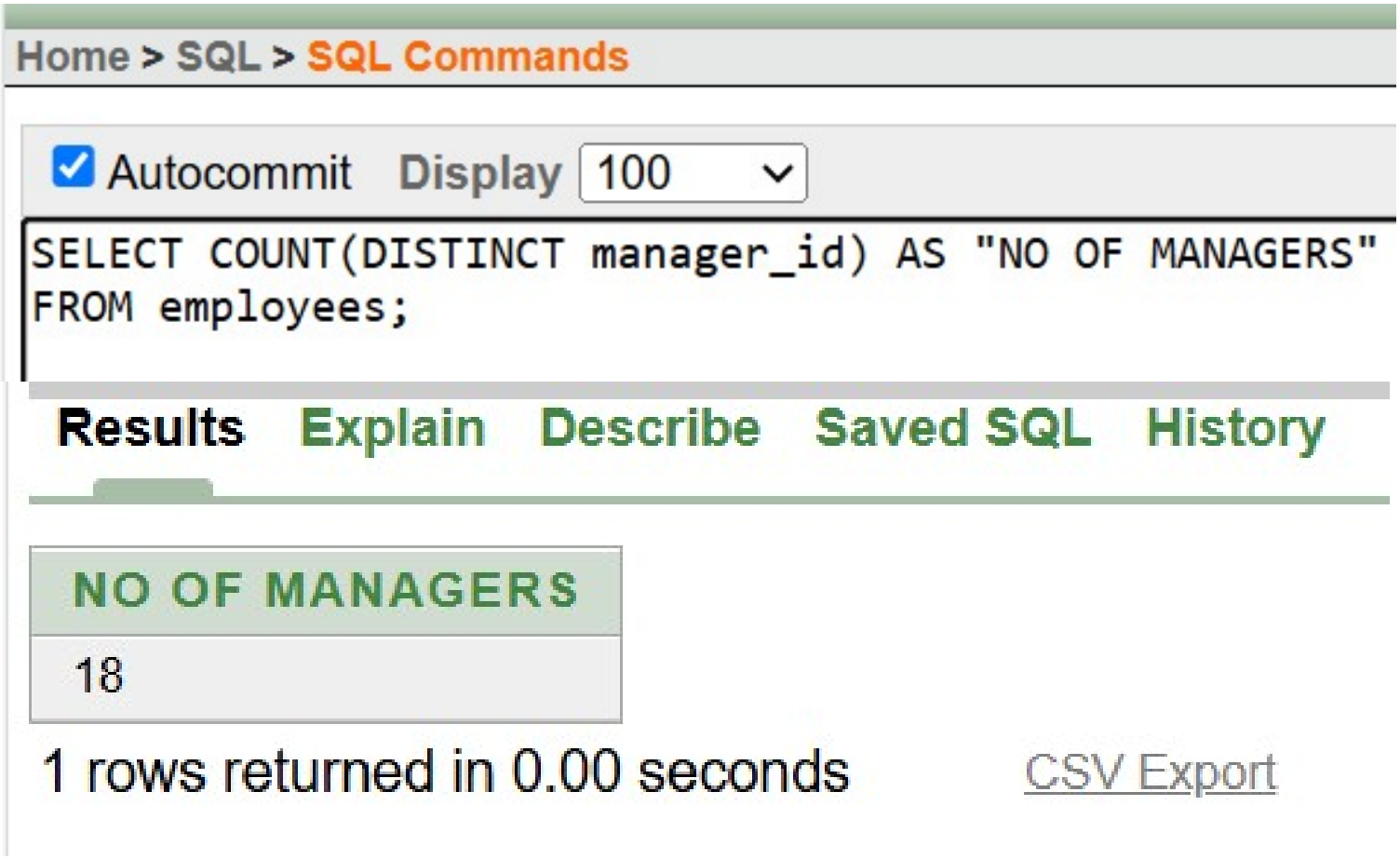
```
SELECT department_id,
       MAX(salary),
       MIN(salary),
       MAX(salary) - MIN(salary) AS "DIFFERENCE"
FROM employees
GROUP BY department_id;
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	MAX(SALARY)	MIN(SALARY)	DIFFERENCE
100	12000	6900	5100
30	11000	2500	8500
-	7000	7000	0
90	24000	17000	7000
20	13000	6000	7000
70	10000	10000	0
110	12000	8300	3700
50	8200	2100	6100
80	14000	6100	7900
40	6500	6500	0
60	9000	4200	4800
10	4400	4400	0

12 rows returned in 0.02 seconds CSV Export

12. **Determine the no of managers without listing them. Label the column (NO OF MANAGERS).**



13. **Write a query to display department_id ,sum of salary ,sum of commission and sum of (salary & commission_amt) for each department . Sort the result department_wise.**

Home > SQL > SQL Commands

Autocommit Display 100

```
SELECT department_id,
       SUM(salary) AS total_salary,
       SUM(NVL(salary * commission_pct, 0)) AS total_commission,
       SUM(salary + NVL(salary * commission_pct, 0)) AS total_salary_commission
FROM employees
GROUP BY department_id
ORDER BY department_id ASC;
```

Results Explain Describe Saved SQL History

DEPARTMENT_ID	TOTAL_SALARY	TOTAL_COMMISSION	TOTAL_SALARY_COMMISSION
10	4400	0	4400
20	19000	0	19000
30	24900	0	24900
40	6500	0	6500
50	156400	0	156400
60	28800	0	28800
70	10000	0	10000
80	304500	72640	377140
90	58000	0	58000
100	51600	0	51600
110	20300	0	20300
-	7000	1050	8050

12 rows returned in 0.00 seconds CSV Export

NOTE:

NVL() is an Oracle function used to replace NULL values with a specified value. If the expression is NULL, NVL returns the replacement value; otherwise, it returns the original value.

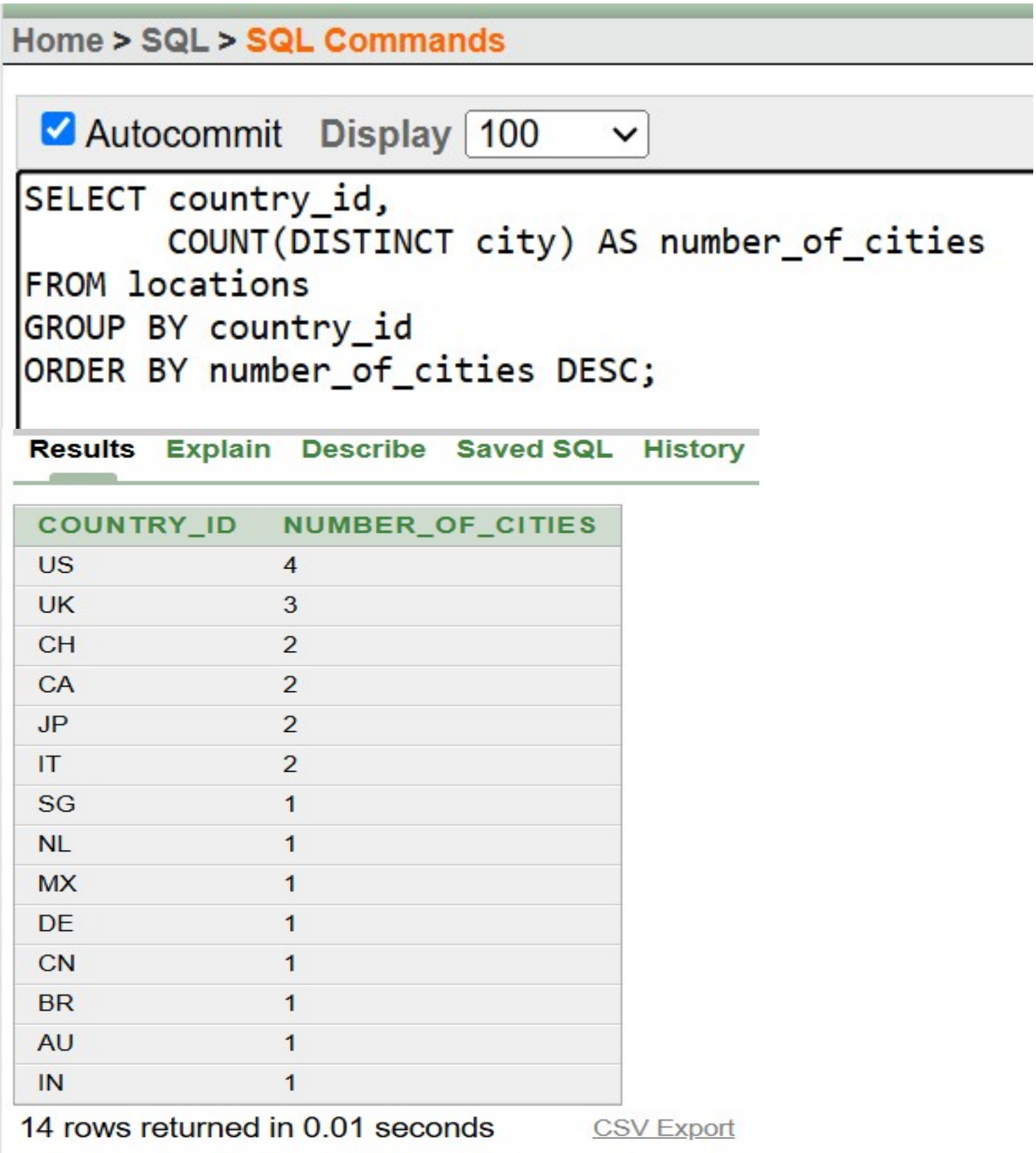
Syntax:

```
NVL(expression, replacement_value)
```

Example:

```
NVL(commission_pct, 0)
```

14. **Write a query to display country-wise number of cities having company offices. List country_id and number of cities and sort as per number of cities in descending order.**



15. **Write a query to display number of employees hired year-wise. Column headings should be : Year and No-of-employees-hired.**

Home > SQL > SQL Commands

Autocommit Display 100

```
SELECT EXTRACT(YEAR FROM hire_date) AS "Year",
       COUNT(*) AS "No-of-employees-hired"
FROM employees
GROUP BY EXTRACT(YEAR FROM hire_date)
ORDER BY "Year" ASC;
```

Results Explain Describe Saved SQL History

Year	No-Of-Employees-Hired
1987	2
1989	1
1990	1
1991	1
1993	1
1994	7
1995	4
1996	10
1997	28
1998	23
1999	18
2000	11

12 rows returned in 0.00 seconds CSV Export

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